

Problem B. A Number Between Two Others

Time Limit 2000 ms

Mem Limit 524288 kB

You are given two integers x and y such that $y > x$ and $y \bmod x = 0$ (that is, y is divisible by x).

Your task is to determine whether there exists an integer z such that

- z lies between x and y (that is, $x < z < y$);
- z is divisible by x (that is, $z \bmod x = 0$);
- y is not divisible by z (that is, $y \bmod z \neq 0$).

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

Each test case consists of a single line containing two integers x and y ($1 \leq x < y \leq 10^{18}$; $y \bmod x = 0$).

Output

For each test case, print the answer as follows: if the required number z exists, print YES; otherwise, print NO. You may print each letter in any case.

Examples

Input	Output
5	NO
1 2	YES
1 3	YES
1234567890 12345678900	YES
2 8	YES
7 84	

Note

In the second test case of the example, you can use $z = 2$.

In the third test case of the example, you can use $z = 7407407340$.